

AI Decision Support for University Ranking and Accreditation^{*}

Preliminary Conceptual Framework

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Abstract This paper presents a preliminary conceptual framework for an AI-based decision support system designed to assist universities in navigating the strategic challenges of ranking and accreditation. Often perceived as externally imposed burdens, these evaluations require collective internal alignment, careful resource allocation, and maintaining academic legitimacy. The proposed framework conceptualizes university decision-making as a multi-dimensional process that must align institutional reality and external evaluation criteria with insider stakeholder interests. To achieve this, we proposed AIDAS-HE, a system that integrates institutional modeling, probabilistic consequence analysis, and the AI-based generation of alternative strategic plans. By leveraging artificial intelligence to explore multiple strategic routes under uncertainty, this framework aims to help university leaders identify actionable plans that improve external metrics while preserving mission-driven value for faculty, students, and staff. Finally, the framework emphasizes the role of AI in translating analytic insights into formal academic communication to ensure institutional legitimacy and consensus.

Keywords: Decision Support Systems, Artificial Intelligence, Higher Education, University Rankings, Academic Accreditation, Institutional Strategy

JEL Codes: I23, C88, D81

1 Background

University ranking and accreditation are often experienced by academic insiders as externally imposed pressures. At first, they may be seen as outsider-defined burdens that consume time, funds, and organizational energy without creating clearly visible rewards for scholars, students, and supporting staff (Hazelkorn 2015). However, once these mechanisms begin to affect enroll-

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ment, reputation, public trust, partnerships, and access to resources, they become unavoidable strategic issues for the university itself.

The challenge is therefore not simply to comply, but to decide well. Universities must mobilize insiders, allocate limited resources, preserve academic legitimacy, and choose action plans that can improve ranking and accreditation outcomes without undermining mission, morale, and long-term capability.

2 Problem Formulation

The core decision problem may be stated as follows:

Given the university's current capabilities, constraints, mission, stakeholder interests, and uncertainty about outcomes, what alternative action plans can most effectively improve ranking and accreditation results while creating acceptable value for insiders?

This problem is difficult because universities are not single-actor organizations. They are collective systems in which multiple parties must act together under formal rules and shared legitimacy.

3 Foundational Ideas

3.1 Decision making as designing the exam

Decision making is not only choosing an answer. It is also formulating the right questions, aligning with value-focused thinking where defining what matters precedes the evaluation of alternatives (Keeney 1992). In a university setting, the institution must ask:

- Which external criteria matter most?
- Where are our strongest and weakest positions?
- Which interventions are feasible with existing resources?
- Which alternatives create the best balance between burden and value?

3.2 Decision making as planning a journey

A university can be viewed as traveling through an evolving institutional landscape:

- **Current position:** present academic and organizational condition
- **Targets:** desired rank and accreditation outcomes
- **Landscape:** regulations, criteria, competitors, public expectations, and internal constraints
- **Routes:** alternative strategic plans
- **Mission:** engine that gives energy and meaning
- **Vision:** horizon that guides direction
- **Competence:** the university's capacity to travel

3.3 Decision making as collective alignment

Because university decisions require participation across many units, collective decision making must be based on shared models of:

- problem reality
- institutional context

- stakeholder interests
- burdens and incentives
- acceptable value exchanges

Consensus becomes possible when the university can generate alternatives that different groups find sufficiently meaningful, fair, and feasible.

3.4 Models for academic legitimacy

Institutional decisions require formal legitimacy to be accepted by the academic community (Suchman 1995). A useful model should be:

1. **Grounded in reality**
2. **Tractable in complexity**
3. **Effective in function**
4. **Clear in communication**

Mathematical, probabilistic, and computational models are especially useful because they help institutions reason systematically, represent uncertainty explicitly, and communicate decisions in academic forms such as reports, proposals, and policy documents.

4 Preliminary Conceptual Framework

4.1 Central proposition

An effective university decision support system should integrate institutional reality modeling, stakeholder interest modeling, probabilistic consequence analysis, and AI-based generation of alternative plans, so that university leaders can choose and communicate action plans that improve ranking and accreditation outcomes while preserving academic legitimacy and insider engagement.

4.2 Main constructs

The framework is built upon several main constructs, as defined and contextualized in Table 1 .

Table 1: Main constructs of the preliminary framework

Construct	Meaning in this framework	Example in university context
Institutional reality	Current internal condition of the university	Faculty profile, student outcomes, research output, governance quality
External evaluation framework	The outsider-defined criteria that shape strategic pressure	Ranking indicators, accreditation standards, compliance requirements
Insider interests	Goals, concerns, and burdens of internal stakeholders	Faculty workload, student learning quality, staff capacity, leadership reputation
Resources and constraints	Means and limits for action	Budget, time, data quality, documentation readiness, digital infrastructure
Alternative plans	Possible routes the university can take	Documentation-first plan, research-strengthening plan, balanced phased plan
Uncertainty	Unknowns affecting outcomes	Methodology changes, weak participation, delayed outputs, policy shifts
Institutional legitimacy	Formal acceptability of decisions	Written reports, senate approval, policy alignment, evidence-based justification
Decision quality	Degree to which the selected plan serves institutional objectives	Better rank, stronger accreditation, manageable burden, mission alignment

4.3 Logical relationships

The framework assumes the following causal logic:

1. **Institutional reality** and **external evaluation frameworks** jointly define the strategic gap.
2. **Stakeholder interests** and **resource constraints** determine which plans are feasible and acceptable.
3. **AI-based alternative generation** expands the range of possible plans beyond ad hoc human responses.
4. **Probability and simulation models** help estimate consequences under uncertainty.
5. **Institutional legitimacy mechanisms** convert analytic outputs into acceptable academic decisions through reports, proposals, and formal deliberation.
6. **Decision quality** improves when the chosen plan produces both external gains and internal value.

4.4 Visual summary

4.5 Framework table

The relationships between these components and their respective phases are summarized in Table 2, detailing the specific modeling needs and expected AI support roles.

Table 2: Framework layers and corresponding AI support roles

Layer	Key question	Modeling need	AI support role	Expected output
Diagnosis	Where are we now?	Institutional baseline model	Data integration, gap detection	Reality profile and weakness map
Interpretation	Why does the gap exist?	Criteria and stakeholder model	Pattern finding, explanation support	Structured problem understanding
Alternative generation	What could we do?	Plan space model	Generate strategic alternatives	Portfolio of candidate plans
Evaluation	What may happen?	Probability and simulation model	Forecasting, scenario analysis	Consequence and trade-off estimates
Legitimization	How do we persuade and align insiders?	Academic report and rhetoric model	Draft reports, briefs, justifications	Formally acceptable proposals
Execution learning	Are we progressing?	Monitoring and adaptation model	Tracking, alerts, recommendation updates	Adaptive implementation guidance

5 Proposed AI Decision Support System

5.1 System purpose

While AI has seen increasing application in higher education, much of it revolves around student learning rather than institutional decision-making (Zawacki-Richter et al. 2019). The proposed system, **AIDAS-HE**, is an AI-based decision support system specifically tailored for helping universities generate, compare, justify, and communicate alternative strategic plans for ranking and accreditation improvement.

5.2 Major modules

5.2.1 1. Institutional Reality Modeling

- University data integration
- Baseline academic and operational profile
- Gap identification

5.2.2 2. Ranking and Accreditation Knowledge Representation

- Criteria and indicator mapping
- Evidence requirement model
- Scoring logic representation

5.2.3 3. Stakeholder Interest and Burden Modeling

- Insider objective mapping
- Burden-benefit analysis
- Possible value-exchange mechanisms

5.2.4 4. Alternative Plan Generation

- Quick compliance alternatives
- Research-centered alternatives
- Student outcome alternatives
- Balanced multi-phase alternatives

5.2.5 5. Consequence Analysis Under Uncertainty

- Scenario analysis
- Probabilistic forecasting
- Multi-criteria comparison

5.2.6 6. Academic Communication and Legitimization

- Briefing notes
- Strategic reports
- Policy drafts
- Accreditation self-evaluation support

6 Illustrative Alternative Plans

6.1 Plan A: Documentation-First Recovery

Focus on evidence completeness, audit readiness, and rapid compliance improvements.

6.2 Plan B: Research Reputation Climb

Focus on publication quality, citation performance, grants, and academic visibility.

6.3 Plan C: Student Outcome Strengthening

Focus on learning quality, employability, student satisfaction, and graduate success indicators.

6.4 Plan D: Balanced Phased Transformation

Combine quick wins, capability development, and long-term institutional strengthening.

7 Researchable Propositions

The preliminary framework suggests several propositions for later empirical study:

1. Universities achieve stronger ranking and accreditation outcomes when external criteria are translated into insider-valued plans rather than treated as pure compliance burdens.
2. AI support improves decision quality when it generates multiple strategic alternatives rather than merely summarizing current performance.
3. Probabilistic consequence analysis improves institutional commitment by making trade-offs and uncertainty more explicit.
4. Academic legitimacy mediates the effect of analytic insight on final institutional action; therefore, report generation and rhetorical framing are not optional add-ons but core functions.
5. Decision support systems are more effective when they help create value exchange among insiders, not merely performance pressure.

8 Implications

8.1 For university leaders

The framework encourages leaders to treat ranking and accreditation as strategic journeys rather than isolated reporting exercises.

8.2 For faculty and staff

The framework highlights the importance of making insider value visible, so institutional participation is not experienced only as extra burden.

8.3 For AI system design

The framework implies that AI should function as an integrator of data, modeling, uncertainty treatment, and formal academic communication.

9 Conclusion

This preliminary conceptual framework positions university ranking and accreditation as collective decision problems requiring realistic models, stakeholder alignment, formal legitimacy, and action under uncertainty. It proposes that AI can support universities most effectively when it helps them diagnose reality, generate alternative plans, estimate consequences, and communicate decisions in forms acceptable within academic institutions.

The framework is preliminary because its constructs, causal relationships, and practical modules still need empirical validation. Nevertheless, it offers a coherent conceptual starting point for developing an AI decision support system that is both analytically rigorous and institutionally appropriate for higher education.

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